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NAJMEH NAYYER

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RESEARCH INTERESTS

Combining experimental and theoretical approaches to elucidate neural principles underlying behavior and cognition, particularly perception and decision-making, with the eventual goal of implementing this knowledge in AI.

EDUCATION

B.Sc., Psychology, University of Isfahan, Isfahan, Iran — Sep 2024 - Present
CGPA: 18.44/20 (3.92/4)

SKILLS

Programming:	Python and MATLAB
Packages:	NumPy, Pandas, SciPy, Matplotlib, PsychoPy, MNE, Scikit-learn
Research Skills:	Statistical Analysis, Experimental Design, Signal Processing, Neural Decoding, Psychophysics
Languages:	English, Persian (National Language)

EXPERIENCE

RESEARCH EXPERIENCE

Research Intern — Oct 2025 - Present
Supervisor: Prof. Alex Clarke, University of Warwick

- Studied how object representations converge and diverge across the brain using ECoG data.

Independent Researcher — Sep 2025 - Present
Supervisor: Prof. Maryam Vaziri-Pashkam, University of Delaware

- Investigated object categorization using object kinematograms.
- Currently fitting psychometric functions and GLMM.

Project Assistant — Apr 2025 - Aug 2025
Supervisor: Prof. Behrad Noudoost, University of Utah — GitHub Link

- Investigated the effect of adaptation on color discrimination along the yellow-blue axis. (see PRESENTATIONS)
- Programmed the paradigm in PsychoPy, contributed to study design, data collection, and analysis pipeline.

MENTORING EXPERIENCE

Study Mentor for the National University Entrance Exam, *Notruphil, Hamedan, Iran* — Mar 2025

- Mentored high school students preparing for the exam and provided academic guidance and support.

PRESENTATIONS

Vahabi, F., Honarmand, F., **Nayyer, N**. (2025). *Chromatic Threshold Stability In The Yellow–Blue Axis*. Poster presented at the Basic and Clinical Neuroscience Congress, Tehran, Tehran, Iran (Link)

SELECTED PROJECTS

Analysis of Natural Object Dataset EEG | Python — Ongoing

- Extracted time-frequency features via Wavelet Transform (Morlet wavelets) in MNE-Python.
- Trained supervised classifiers to decode object categories across neural representations.

Bootcamp Projects | Python (GitHub Link) — 2026

- Developed 10+ interactive Python applications spanning games, GUIs, and utility systems using OOP, event-driven programming, and data management to strengthen core coding skills for computational research.

Gender Inequality Attitudes Questionnaire | SPSS — Jul 2026

- Co-designed and validated a questionnaire with excellent reliability (Spearman-Brown = 0.978).
- Calculated descriptive statistics, standardized Z/T scores, and multiple cut-off score and reliability measures.

Memory-Guided Saccade Task | MATLAB (Project Link) — Jun 2026

- Visualized spike trains and LFP from macaque FEF during a memory-guided saccade task. (8 angles, 2 radii)
- Performed angle and radius encoding (AUC and MI) across take phases and target location decoding (SVM).

Analysis of Psychology Class of 2028 Socioeconomic Status | Python (GitHub Link) Aug 2025

- Preprocessed and analyzed data using methods such as repeated measures ANOVA and premutation tests.

AWARDS & EXTRACURRICULAR

Brainies Book Club Founder and Leader (Link) Jan 2026 - Present

- Presented “A Thousand Brains: A New Theory of Intelligence” by Jeff Hawkins to +10 students.

Vice President, Psychology students' association of University of Isfahan, Isfahan, Iran Jun 2025 - Present

Top 0.003% Nationwide, Iranian National University Entrance 2024

- Ranked 1,135 / 348,900+ candidates and awarded undergraduate tuition scholarship.

SELECTED OPTIONAL COURSES

Mathematics for Machine Learning: Multivariate Calculus, Samuel J. Cooper, ICL (Coursera) Ongoing
NeuroCONNECT Summer School (Inspired by Neuromatch), University of Tehran (Online) Aug 2026
Supervised Machine Learning: Regression and Classification, Andrew Ng, DeepLearning.AI (Coursera) Nov 2025
Understanding and Visualizing Data with Python, Brenda Gunderson, University of Michigan (Coursera) Aug 2025
Principles of Neuroscience, Behrad Noudoost, Isfahan University of Medical Sciences May 2025
Data Structures, Charles Severance, University of Michigan (Coursera) May 2025
Introduction to statistics, Guenther Walther, Stanford (Coursera) Mar 2025
AI for Everyone, Andrew Ng, DeepLearning.AI (Coursera) Jan 2025
Understanding the Brain: Neurobiology of Everyday Life, Peggy Mason, University of Chicago (Coursera) Dec 2024
Introduction to Data Analysis Using Excel, Sharad Borle, Rice University (Coursera) Nov 2024